

Lecture 3:

Quick Intro to Classification and Log-linear Models

Reminder: Supervised Learning

- The task:
 - **Input:** training samples $x_1, x_2, x_3, \dots, x_n$ drawn from some distribution D , the learner is provided with their labels $y_1, \dots, y_n$
 - **Goal:** correctly predict the label of a new sample drawn from D
- Evaluation:
 - Take an annotated corpus and partition it as follows:



- Development data → for exploration; test data → for reporting results

Probabilistic Classification

- Two broad approaches to predicting classes y^*
 1. Joint / Generative: estimate a joint probabilistic model of the data
 - Usually assumes a functional form for $P(x/y)$, $P(y)$ and estimate parameters from the data
 - Use Bayes rule to compute $P(y/x)$
 - Inference:

$$y^* = \operatorname{argmax}_y P(y|x) = \operatorname{argmax}_y \frac{P(y)P(x|y)}{P(x)} = \operatorname{argmax}_y P(y)P(x|y)$$

- Advantages: learning is generally easy, smoothing is well-understood, a complete model of the data

Probabilistic Classification

2. Conditional / Discriminative (e.g., Logistic Regression)

- Only model conditional probability $P(y/x)$
- Prediction can be done directly:

$$y^* = \operatorname{argmax}_y P(y/x)$$

- Advantages: no need to model $P(x)$, easier to develop feature-rich models for $P(y/x)$